AIS Tanker Tracking System

Abstract

List of Abbreviations

1.0 Introduction

Automatic Identification System (AIS) data is widely used in the maritime domain to monitor vessel movements and support navigation safety. AIS messages contain useful information such as vessel position, speed, heading, destination, and navigation status. According to the International Maritime Organization, AIS is used to automatically provide ship information to other ships and coastal authorities, which makes it an important source of maritime traffic data. (1)

1.1 Background and Motivation

Maritime transportation plays an important role in global trade, logistics, and coastal security. Every day, many vessels travel through busy sea routes, including cargo ships, passenger vessels, and tankers. Monitoring these movements is important because it helps improve maritime safety, traffic awareness, and decision-making.

AIS data is valuable because it allows vessel movements to be studied over time. Previous research has shown that AIS data can be used for route estimation, anomaly detection, collision prediction, and path planning. This shows that AIS is not only useful for showing the current location of a vessel, but also for understanding vessel behaviour and movement patterns.(2)

However, in practical work, AIS data is not always clean or easy to use. Large AIS datasets may contain missing values, duplicate records, invalid coordinates, inconsistent timestamps, and incomplete destination information. Because of this, raw AIS data normally needs to be cleaned, filtered, and stored properly before it can be used for analysis.

This project is motivated by the need to build a structured system that can process AIS data in a reliable way. Instead of only focusing on prediction or visualization, the project focuses on creating the backend foundation needed for maritime analytics. This includes data preprocessing, tanker filtering, database storage, API support, and future machine learning integration.

1.2 Problem Statement

The maritime industry generates a massive amount of Automatic Identification System (AIS) data through continuous vessel transmissions containing information such as location, speed, heading, and voyage details. Managing and analyzing this large-scale maritime data manually is difficult and requires automated processing techniques for effective maritime situational awareness and vessel monitoring [1]. Raw AIS datasets often contain duplicate records, missing values, invalid coordinates, inconsistent timestamps, and noisy information, making direct analysis and visualization challenging.

In addition, most existing maritime systems primarily focus on route prediction and anomaly detection, while limited attention is given to developing scalable backend infrastructures capable of handling real-world AIS datasets efficiently. The absence of structured tanker-specific AIS processing platforms creates difficulties in storing, preprocessing, filtering, and visualizing maritime information in a reliable and scalable manner.

Therefore, there is a need for a scalable AIS tanker analytics system capable of automatically extracting, cleaning, filtering, storing, and visualizing tanker vessel data. The system should support ETL-based preprocessing, database integration, REST API accessibility, Docker-based deployment, and web-based maritime visualization while also providing a foundation for future machine learning and explainable AI applications.

1.3 Research Objective

1.4 Project Scope and Limitation

2.0 Literature Review and Theories

2.1 Summary of prior studies:

Study 1

Pallotta et al. proposed the TREAD (Traffic Route Extraction and Anomaly Detection) framework for extracting maritime traffic knowledge from AIS data using an unsupervised and incremental learning approach. The framework was designed to identify vessel movement patterns, detect anomalous maritime behavior, and predict future vessel routes using AIS information collected from terrestrial and satellite receivers. The study demonstrated how large-scale AIS data could be transformed into meaningful maritime situational awareness through automated processing techniques.(3)

The TREAD framework mainly focused on traffic route extraction, anomaly detection, and route prediction using statistical and clustering-based methods such as DBSCAN. In contrast, the current project focuses specifically on AIS tanker vessel data processing and infrastructure development. The proposed system emphasizes scalable ETL processing, tanker data filtering, database management using PostgreSQL, Docker-based deployment, REST API integration, and web-based maritime visualization. While the TREAD framework concentrates on intelligent maritime behavior analysis, the present project builds a structured backend architecture capable of supporting future maritime analytics, machine learning, and explainable AI applications.

2.2 Python

2.3 Web-development

2.4 Project related theory

3.0 Methodology overview

3.1 System Architecture

3.2 Research Question

4.0 Dataset

4.1 Data Description and Explanation

The dataset used in this project is based on AIS (Automatic Identification System) data collected from maritime vessels. AIS is a tracking system used in ships to continuously transmit important navigational and identification details for maritime safety and monitoring purposes. This data helps in tracking vessel movement, monitoring maritime traffic, and improving navigation management.

The AIS dataset contains both static and dynamic vessel information generated during ship movement. One of the important fields in the dataset is MMSI (Maritime Mobile Service Identity), which is a unique nine-digit number assigned to every vessel for identification and communication purposes. Some records also include the IMO (International Maritime Organization) number, which acts as a permanent international identification number for ships.

The dataset also contains positional information such as Latitude and Longitude, which represent the real-time location of vessels on the sea. Vessel movement information is represented using SOG (Speed Over Ground), which indicates the speed of the vessel in knots, and COG (Course Over Ground), which shows the direction in which the vessel is moving. Another important attribute is Heading, which represents the actual direction towards which the front side of the ship is pointing.

Each AIS message also contains a Timestamp field that records the exact date and time when the information was transmitted. Additional information such as Navigation Status and Destination is also available in the dataset. Navigation Status describes the current condition of the vessel, such as whether the ship is moving, anchored, or stationary, while the Destination field contains the intended arrival location entered by the vessel operator. The Vessel Type attribute was used to filter tanker vessels specifically for this project.

Since AIS data is generated continuously by ships, the raw dataset is very large and contains several real-world data quality issues such as duplicate records, missing values, inconsistent timestamps, and invalid coordinates. Therefore, preprocessing and validation steps were performed before storing the data in the PostgreSQL database. The cleaned and processed tanker AIS dataset is then used for vessel tracking, maritime visualization, and future analytics applications such as anomaly detection, route prediction, and machine learning-based maritime intelligence systems.

4.2 Data preprocessing

Before the AIS data could be used for storage and analysis, several preprocessing and validation steps were performed. Raw AIS data often contains inconsistencies such as missing values, invalid coordinates, duplicate records, malformed timestamps, and incomplete vessel identifiers. Therefore, a structured ETL (Extract, Transform, Load) pipeline was implemented using PostgreSQL and SQL transformation scripts to clean and organize the dataset.

Initially, all incoming AIS records were loaded into a staging table (`tanker_staging`). This table stores the raw AIS messages exactly as they are received from the source files. Since many AIS attributes are transmitted as text values, most fields in the staging layer were stored in raw string format before conversion and validation. This staging approach allowed raw source data to be preserved before applying transformations.

During preprocessing, tanker vessels were filtered using the AIS `ship_type` field. Only records classified as “Tanker” were processed further in the pipeline. This filtering step reduced unnecessary storage usage and improved processing efficiency by excluding unrelated vessel categories.

Several data cleaning operations were then applied. Raw timestamp values were converted into structured UTC timestamps using SQL timestamp conversion functions. Positional attributes such as latitude and longitude were transformed from raw text values into numeric floating-point values suitable for spatial analysis. Coordinate validation checks were also performed to ensure that latitude values remained within the valid range of -90 to 90 degrees and longitude values remained within the range of -180 to 180 degrees.

Additional navigational attributes including SOG (Speed Over Ground), COG (Course Over Ground), ROT (Rate of Turn), heading, and draught were converted into numeric formats using SQL casting operations. Empty strings and malformed numeric values were converted into NULL values to improve consistency in the final database tables.

The preprocessing stage also included validation of vessel identifiers. IMO values were cleaned using trimming operations and validated using regular expression checks to ensure proper formatting. MMSI identifiers were additionally validated using MID (Maritime Identification Digits) country code lookup tables. Vessel flag information was derived from MMSI country prefixes using a lookup-based mapping table.

Records containing missing or unknown IMO values were not discarded completely. Instead, these records were inserted separately into the `tanker_positions` table and flagged as anomalous records using anomaly tracking fields. Additional anomaly categories such as invalid MMSI values, invalid positional coordinates, missing draught values, and missing destination values were also identified during preprocessing. Dedicated anomaly tracking tables (`anomaly_types` and `anomaly_flags`) were implemented to support future anomaly analysis and human-in-the-loop validation workflows.

Duplicate AIS position records were prevented using unique constraints defined on tanker ID, timestamp, latitude, and longitude values. Database indexes were also created on frequently queried columns such as IMO, MMSI, timestamps, and tanker identifiers to improve query performance on large AIS datasets.

After preprocessing and validation, cleaned vessel metadata was stored in the `tankers` table, while cleaned vessel movement data was inserted into the `tanker_positions` table. The processed dataset now provides a structured foundation for future maritime analytics tasks such as voyage detection, anomaly detection, route analysis, and machine learning applications.

5.0 System Integration and web application

5.1 API development

5.2 Frontend development

5.3 Integration challenges

6.0 Result and Analysis

7.0 Discussion

8.0 Future Work

9.0 Conclusion

References:

1)https://www.imo.org/en/ourwork/safety/pages/ais.aspx?utm_source

2) https://arxiv.org/pdf/1606.00981

Acknowledgement

Appendices